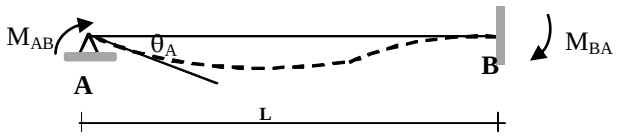
Moment Distribution Method (Hardy Cross)

The cross method of moment distribution is based up on the same basic assumptions made in the slope deflection method where the end moment of members are considered to be the superimposed effect of fixed end moments and end moments due to rotations and relative displacement of the ends. In the slope deflection method, the final end moments are obtained after obtaining the rotations and relative displacements by solving a set of linear simultaneous equations formulated by considering equilibrium condition of joints, members or the whole structure. Unlike this, in cross method of moment distribution, the final member end moments can directly be obtained by correcting /adjusting the fixed end moments through balancing /distribution of the unbalanced moment at a joint to members framing at the joint in proportion to their stiffness through successive iteration until satisfactorily accurate results are obtained. Before going to the development and application of the method, a review on the following fundamental relationships is found to be important.

(i) **Stiffness Factor:** It is found out that



$$M_{AB} = \frac{4EI}{L}\theta_A = 4EK\theta_A$$

$$M_{BA} = 2EK\theta_A$$

$$M_{AB}/\theta_A = 4EK$$

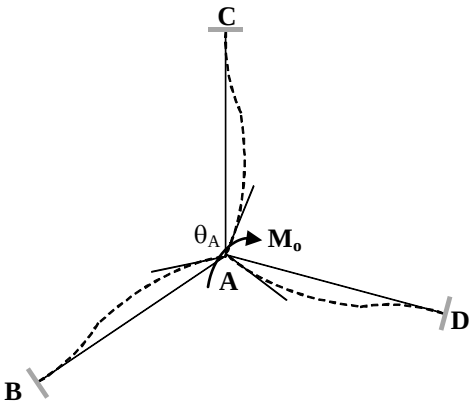
The value M_{AB}/θ_A is a measure of the resistance of the beam for rotation and hence called stiffness of the member.

(ii) Cary Over Factor

$$M_{BA}/M_{AB} = \frac{2EK\theta_A}{4EK\theta_A} = 1/2$$

The ratio of the induced moment at the far end M_{BA} to the applied moment at the near end M_{AB} is defined as the carry over factor.

(iii) Distribution Factor:



The unbalanced moment at a joint is Distributed in to the members framing at the joint in proportion to their stiffness.

$$M_{AB} = 4EK_{AB}\theta_A$$

$$M_{AC} = 4EK_{AC}\theta_A$$

$$M_{AD} = 4EK_{AD}\theta_A$$

From Equilibrium condition:

$$M_{AB} + M_{AC} + M_{AD} = M_o$$

$$4EK_{AB}\theta_A + 4EK_{AC}\theta_A + 4EK_{AD}\theta_A = M_o$$

$$4E\theta_A\left(K_{AB} + K_{AC} + K_{AC}\right) = M_o$$

$$4E\theta_A = \frac{M_o}{\sum K}$$

Therefore the end moments are expressed as:

$$M_{AB} = 4EK_{AB}\theta_A = \frac{K_{AB}}{\sum K}M_o = \left(DF\right)_{AB}M_o$$

$$M_{AC} = 4EK_{AC}\theta_A = \frac{K_{AC}}{\sum K}M_o = \left(DF\right)_{AC}M_o$$

$$M_{AD} = 4EK_{AD}\theta_A = \frac{K_{AD}}{\sum K}M_o = \left(DF\right)_{AD}M_o$$

The term $K/\sum K$ is known as the distribution factor.

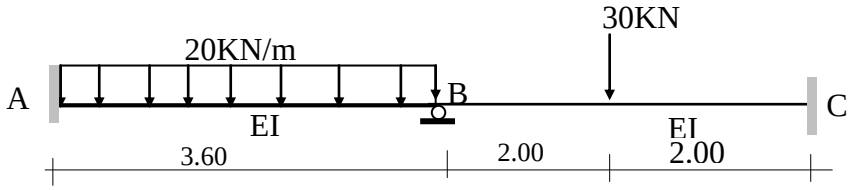
Application of Moment Distribution

(i) **Continuous Beams with Fixed Ends:**

The fixed end moments are calculated in each span. The unbalanced fixed end moments at the joints or intermediate supports are then adjusted /corrected by distributing the unbalanced moment to each of the span through iterative scheme. Distribution factor of zero is assigned at the fixed end due to the fact that the fixed support is infinitely rigid as compared to stiffness of the beam. The procedure can be explained through the following example.

Example:

Determine the support moments for the following beam



Soln.

Fixed End Moments: $M_{AB}^F = -M_{BA}^F = \frac{wL^2}{12} = \frac{20*3.6^2}{12} = 21.60$

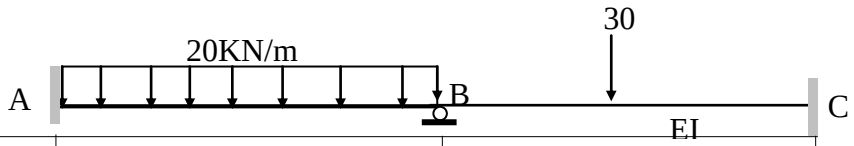
$M_{BC}^F = -M_{CB}^F = \frac{PL}{8} = \frac{30*4}{8} = 15.00$

Stiffness (I = 3.6): $k_{AB} = \frac{3.6}{3.6} = 1;$ $k_{BC} = \frac{3.6}{4} = 0.9$

Distribution factors:

$DF_{BA} = \frac{1}{1+0.9} = 0.526;$ $DF_{BC} = \frac{0.9}{1+0.9} = 0.474$

$DF_{AB} = 0$; $DF_{CB} = 0$



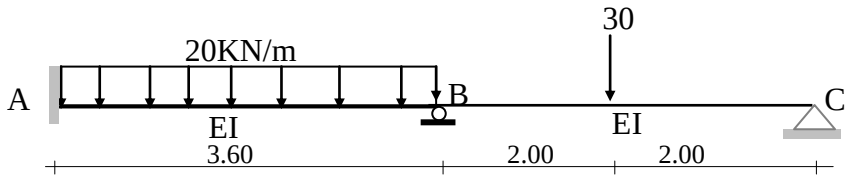
K	1.0		0.90	
DF	0	0.526	0.474	0
FEM	+21.60	-21.60	+15.00	-15.00
	+1.74	+3.47	+3.13	+1.57
Final Moment	+23.34	+18.13	-18.13	-13.43

(ii) Beams having hinged ends:

The same procedure of distributing unbalanced moments can be carried out with the inclusion of DF = 1.0 at the hinged end instead of 0.0 since the hinge support doesn't absorb any moment , and the iteration is continued until the final moment at the hinged end becomes zero.

Example:

Repeat the above example with end C being hinged.

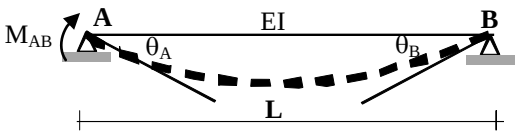


K	1.0		0.90	
DF	0	0.526	0.474	1
FEM	+21.60	-21.60	+15.00	-15.00
	+1.74	+3.47	+3.13	+1.57
		+6.72		+13.43
	-1.76	-3.53	-3.18	-1.59
		+0.80		+1.59
	-0.21	-0.42	-0.38	-0.19
		+0.09		+0.19
	-0.03	-0.05	-0.04	-0.02
				+0.02
Final Moment	+21.34	-22.14	+22.14	0

This lengthy iteration can considerably be shortened by using the appropriate stiffness value for segment BC of the beam where the effect of hinge support at C is taken in to account initially and the fixed end moment is determined accordingly so that no carry over is made to the hinged end.

Stiffness value when the far end is hinged instead of fixed.

From SD equation:



$$M_{BA} = -2EK(2\theta_B + \theta_A) = 0$$

$$\Rightarrow \theta_B = -\theta_A/2$$

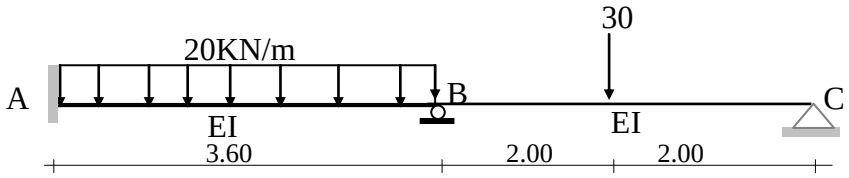
$$M_{AB} = -2EK(2\theta_A + \theta_B)$$

$$\Rightarrow M_{AB} = 3EK\theta_A = \frac{3}{4}(4EK\theta_A)$$

Therefore the stiffness for a span of a beam when the far end is hinged is found to be three-fourth of the stiffness for the case of both ends fixed. Cantilevers are considered to have zero stiffness.

Considering the previous example, the stiffness for span BC is taken to be ¾ of I/L and the fixed end moment is determined as propped cantilever.

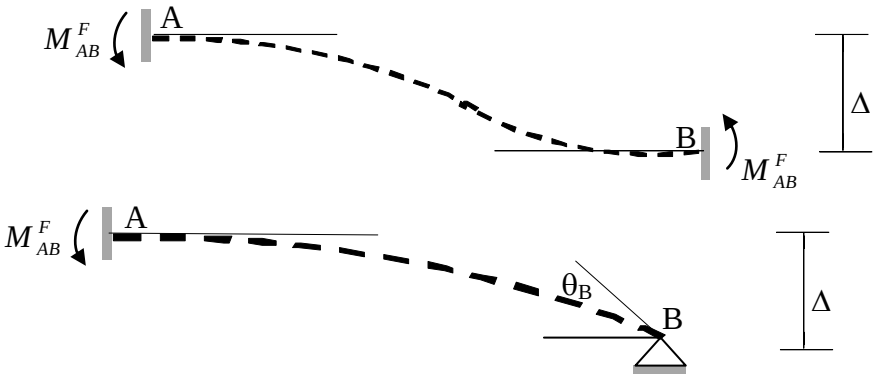
$$K_{BC} = \frac{3}{4}\left(\frac{I}{L}\right) = \frac{3}{4}\left(\frac{3.6}{4}\right) = 0.68 \quad ; \quad M^F_{BC} = \frac{3PL}{16} = \frac{3*30*4}{16} = 22.50$$



K	1.0		0.68	
DF	0	0.60	0.40	
FEM	+21.60	-21.60	+22.50	
		-0.54	-0.36	
	-0.27			
Final Moment	+21.33	-22.14	+22.14	0

(iii) Continuous Beams with support Settlements

The moment distribution can also be used to analyze continuous beams with support settlement and rotation by incorporating the moments due to prescribed rotation and displacement of supports as equivalent fixed end moments, and then the normal distribution /balancing of moment can be followed.



$$M_{AB}^{F} = M_{BA}^{F} = \frac{6EI}{L^2}\Delta$$

$$M_{AB}^{F} = \frac{3EI}{L^2}\Delta \;;\; M_{BA}^{F} = 0$$

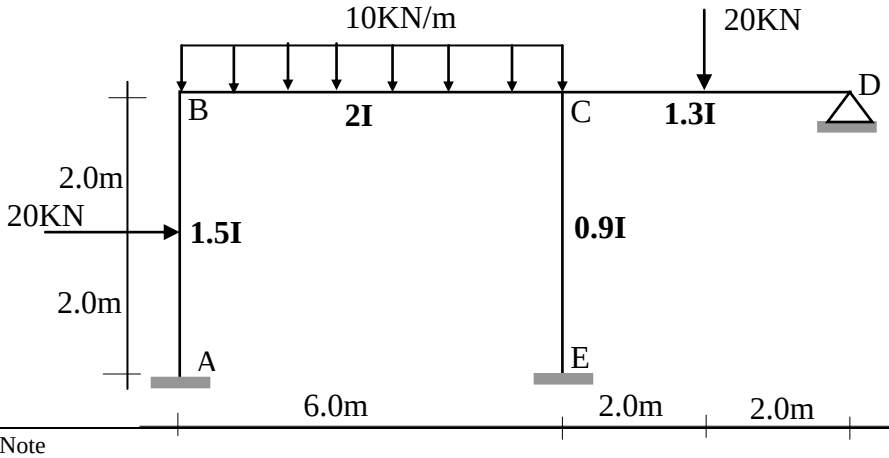
$$\theta_B = \frac{3\Delta}{2L}$$

(iv) Frames With Out Side Sway

Frames with out side sway can be analyzed in the same manner as that of the continuous beam. The only difference would probably be, in the case of frames there may be more than two members framing at a joint. The unbalanced moment at a joint is still distributed to each member in proportion to their stiffness.

Example:

Determine the moments at the joints for the following frame using cross method of moment distribution.



Solution:

- Relative stiffness (let I = 12):

$$K_{AB} = \frac{1.5}{4} * 12 = 4.5 \quad ; \quad K_{BC} = \frac{2}{6} * 12 = 4$$

$$K_{CD} = \frac{3}{4} * \frac{1.3}{4} * 12 = 2.93 \quad ; \quad K_{CE} = \frac{0.9}{4} * 12 = 2.7$$

- Distribution factor, DF:

$$DF_{BA} = \frac{K_{BA}}{K_{BC} + K_{BA}} = \frac{4.5}{4 + 4.5} = 0.53 \quad ; \quad DF_{BC} = \frac{4}{4 + 4.5} = 0.47$$

$$DF_{CB} = \frac{K_{CB}}{K_{CB} + K_{CD} + K_{CE}} = \frac{4}{4 + 2.93 + 2.7} = 0.42 \quad ; \quad DF_{CD} = \frac{2.93}{4 + 2.93 + 2.7} = 0.3$$

$$DF_{CE} = \frac{2.7}{4 + 2.92 + 2.7} = 0.28 \quad ; \quad DF_{AB} = DF_{EC} = 0$$

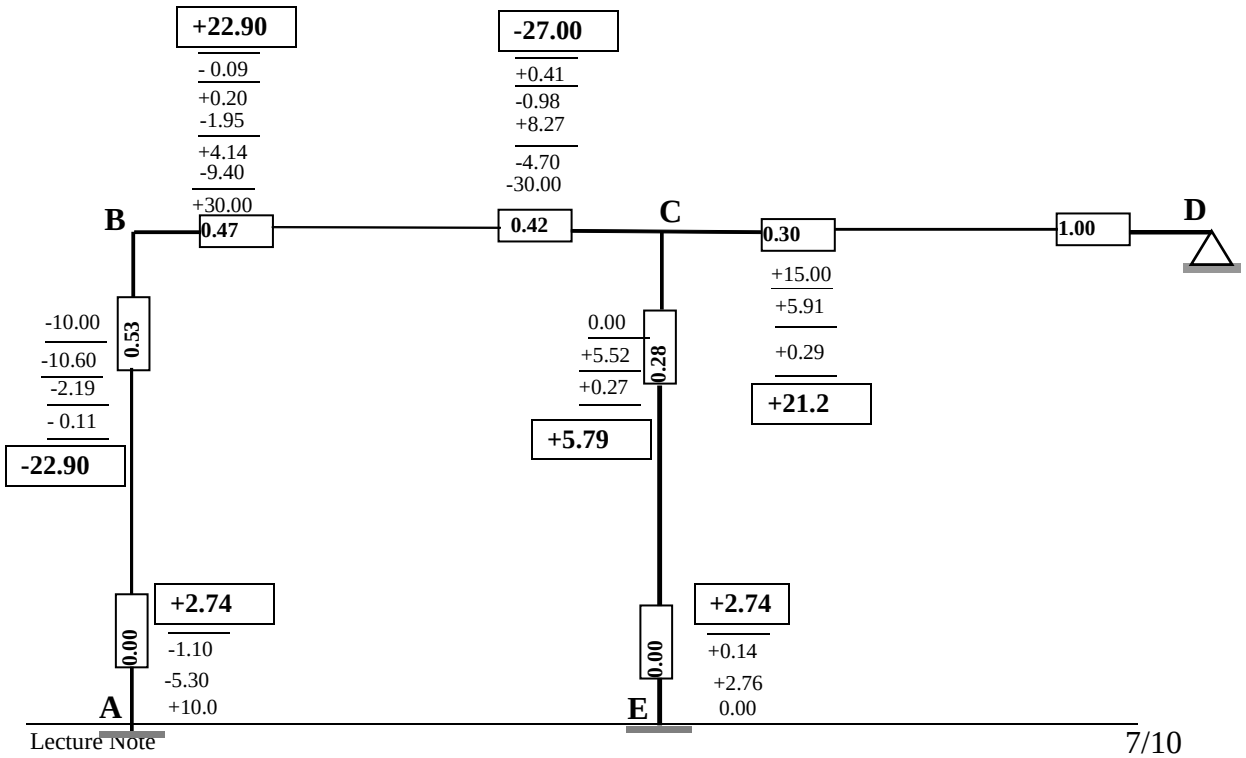
- Fixed end moment:

$$M^{F}_{AB} = -M^{F}_{BA} = \frac{20 * 4}{8} = 10\ KN.m$$

$$M^{F}_{BC} = -M^{F}_{CB} = \frac{10 * 6^2}{12} = 30\ KN.m$$

$$M^{F}_{CD} = \frac{3 * 20 * 4}{16} = 15\ KN.m$$

End moments of members are obtained by balancing the fixed end moments iteratively as shown below with the first adjustment of moments is started at joint B



(v) Frame with Side Sway:

If the frame is subjected to lateral loads, or unsymmetrical load or the frame itself is unsymmetrical, in addition to rotation, the frame undergoes side sway. As the joint displacements are unknown, the fixed end moments due to displacements can not be calculated initially and included in the iteration. Therefore moment distribution method for such cases can be applied by dividing the problem in to two stage of solution:

Consider the frame shown in fig (a) below. The joints are first assumed to be held against side sway, by introducing artificial restraint R, at the appropriate joint as shown in fig (b) and the end moments corresponding to this case are determined through normal balancing/distribution procedure. The corresponding reaction R at the artificial joint is determined from equilibrium consideration of the frame.

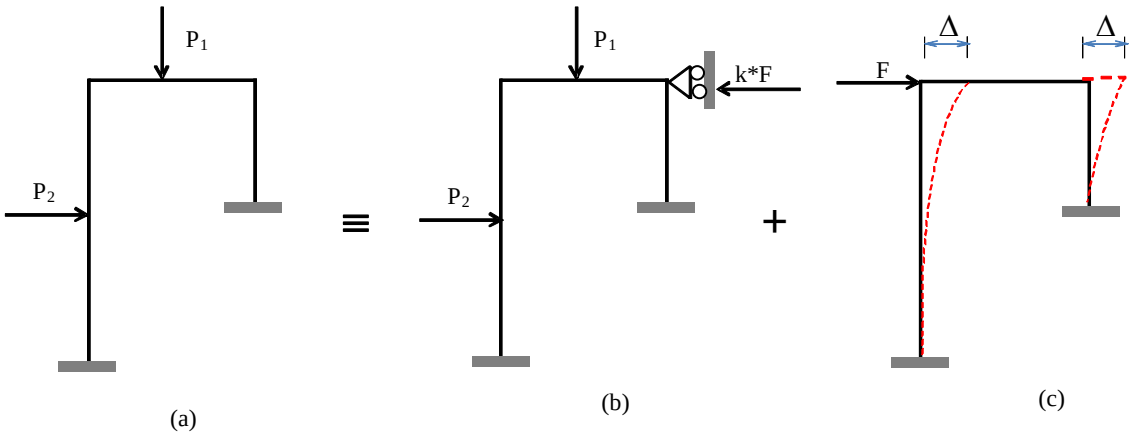
The artificial restraint is released and the frame is assumed to undergo a sideway (some reasonable value for the displacement can be considered) as shown inn fig (c) in the absence of the applied loads. The fixed end moments due to the displacement are determined and distributed by moment distribution method. The equivalent force F necessary to maintain the frame in its swayed position corresponding to the specified displacement Δ is determined from equilibrium condition of the frame.

The actual magnitude of the lateral force F necessary to eliminate the artificial restrain R is:

R= kF, where k is a correction factor

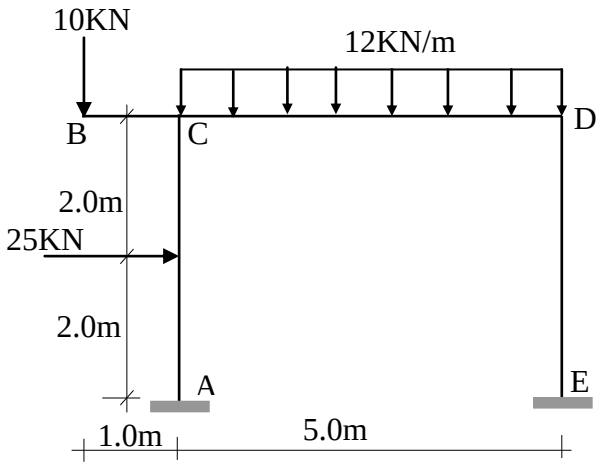
The final set of end moments is obtained by adding the non-sway moments to the sway moments multiplied by the correction factor.

$$M_a = M_b + kM_c$$



Example:

Determine the support moment using moment distribution and draw the shear force and bending moment diagrams for the following frame.



EI – constant for all members

Exercise:

Using moment distribution, determine the end moments and draw the shear force and bending moment diagrams for the following frames.

